

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Data Visualization with Power BI and Tableau

Practical – II

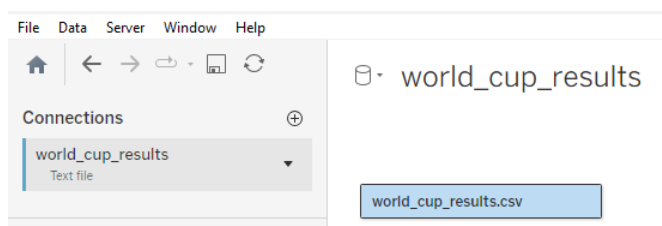
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Class: TYIT	Batch: 01
Date of Assignment: 31-07-2026	Date/Time of Submission: 31-07-2026

Working with Data Sources and Data Preparation in Tableau

a) Connect to different data sources (Excel, CSV, Database)

Steps:

1. Open Tableau Desktop.
2. On the Start page, select the required data source:
 - Microsoft Excel
 - Text File (CSV)
 - Database (MySQL, SQL Server, Oracle, etc.)
3. Browse and select the file or enter database credentials.
4. Load the data into Tableau.



b) Create and use data extracts vs live connections

Steps:

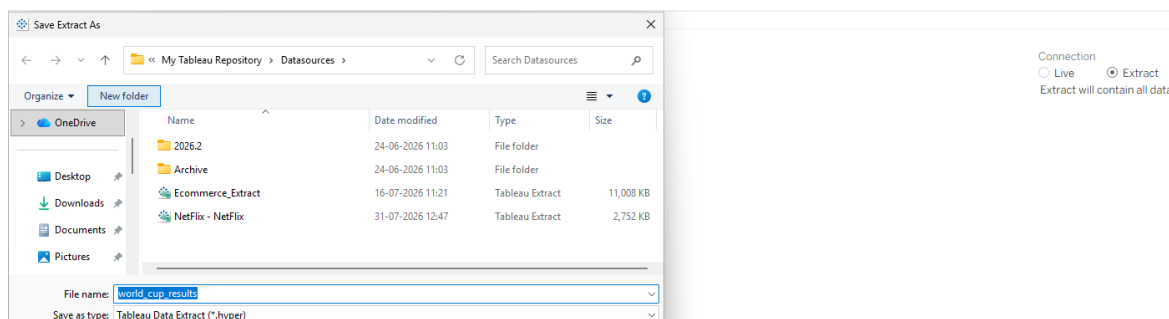
After connecting to data, choose either:

- **Live connection**
- **Extract**

For an Extract:

- Click **Extract**.
- Apply filters if needed.
- Save the extract (.hyper).

Use the extract in your workbook.

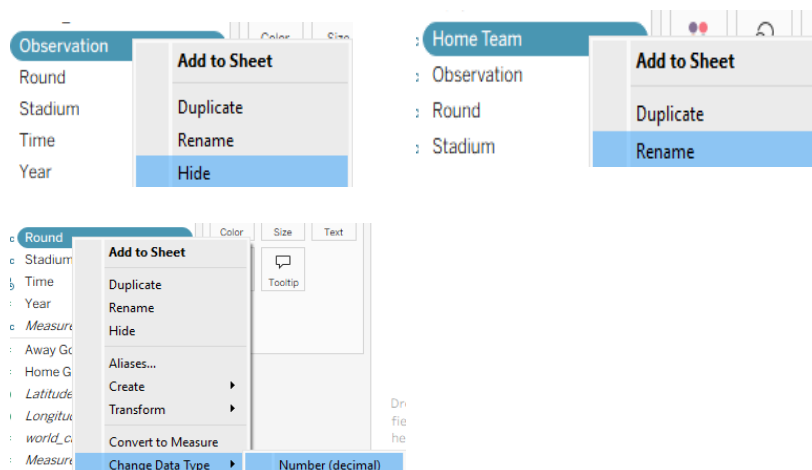


c) Manage metadata (rename, hide, change data types)

Steps:

1. Go to the **Data Source** page.
2. Rename fields by double-clicking their names.
3. Hide unused columns using **Hide**.

4. Change data types by clicking the data type icon beside each field.



d) Perform joins (inner, left, right) on multiple tables

Steps:

1. Connect to the first table.
2. Drag another table onto the canvas.
3. Select the join key.
4. Choose the join type:
 - Inner Join
 - Left Join
 - Right Join
5. Verify the joined data.

Join ✕


Inner


Left


Right


Full Outer

Data Source		NetFlix - NetFlix.csv	
Country	=	country	
Add new join clause			

# world_cup_results.csv Year	 world_cup_results.csv Date	 world_cup_results.csv Time	Abc world_cup_results.csv Round	Abc world_cup_results.csv Stadium
2014	12-06-2014	30-12-1899 17:00:00	Group A	Arena de Sao Paulo
2014	13-06-2014	30-12-1899 13:00:00	Group A	Estadio das Dunas
2014	13-06-2014	30-12-1899 16:00:00	Group B	Arena Fonte Nova
2014	13-06-2014	30-12-1899 18:00:00	Group B	Arena Pantanal
2014	14-06-2014	30-12-1899 13:00:00	Group C	Estadio Mineirao
2014	14-06-2014	30-12-1899 16:00:00	Group D	Estadio Castelao

Join



Inner



Left



Right



Full Outer

Data Source

NetFlix - NetFlix.csv

Country

=

country

Add new join clause

100

→

#	world_cup_results.csv	world_cup_results.csv	world_cup_results.csv	Abc world_cup_results.csv	Abc world_cup_results.csv	🌐 world_cup_results.csv
Year	Date	Time	Round	Stadium	City	
2014	12-06-2014	30-12-1899 17:00:00	Group A	Arena de Sao Paulo	Sao Paulo	
2014	13-06-2014	30-12-1899 13:00:00	Group A	Estadio das Dunas	Natal	
2014	13-06-2014	30-12-1899 16:00:00	Group B	Arena Fonte Nova	Salvador	
2014	13-06-2014	30-12-1899 18:00:00	Group B	Arena Pantanal	Cuiaba	
2014	14-06-2014	30-12-1899 13:00:00	Group C	Estadio Mineirao	Belo Horizonte	
2014	14-06-2014	30-12-1899 16:00:00	Group D	Estadio Castelao	Fortaleza	

The screenshot shows a 'Join' dialog box with four join types: Inner, Left, Right, and Full Outer. The 'Right' join type is selected. Below the dialog, a data table is displayed with columns: #, world_cup_results.csv, world_cup_results.csv, world_cup_results.csv, world_cup_results.csv, world_cup_results.csv. The table contains 6 rows of data.

#	world_cup_results.csv	world_cup_results.csv	world_cup_results.csv	world_cup_results.csv	world_cup_results.csv
Year	Date	Time	Round	Stadium	
2014	12-06-2014	30-12-1899 17:00:00	Group A	Arena de Sao Paulo	
2014	13-06-2014	30-12-1899 13:00:00	Group A	Estadio das Dunas	
2014	13-06-2014	30-12-1899 16:00:00	Group B	Arena Fonte Nova	
2014	13-06-2014	30-12-1899 18:00:00	Group B	Arena Pantanal	
2014	14-06-2014	30-12-1899 13:00:00	Group C	Estadio Mineirao	
2014	14-06-2014	30-12-1899 16:00:00	Group D	Estadio Castelao	

The screenshot shows a 'Join' dialog box with four join types: Inner, Left, Right, and Full Outer. The 'Right' join type is selected. Below the dialog, a data table is displayed with columns: #, world_cup_results.csv, world_cup_results.csv, world_cup_results.csv, world_cup_results.csv, world_cup_results.csv. The table contains 6 rows of data.

#	world_cup_results.csv	world_cup_results.csv	world_cup_results.csv	world_cup_results.csv	world_cup_results.csv
Year	Date	Time	Round	Stadium	City
2014	12-06-2014	30-12-1899 17:00:00	Group A	Arena de Sao Paulo	Sao Paulo
2014	13-06-2014	30-12-1899 13:00:00	Group A	Estadio das Dunas	Natal
2014	13-06-2014	30-12-1899 16:00:00	Group B	Arena Fonte Nova	Salvador
2014	13-06-2014	30-12-1899 18:00:00	Group B	Arena Pantanal	Cuiaba
2014	14-06-2014	30-12-1899 13:00:00	Group C	Estadio Mineirao	Belo Horizonte
2014	14-06-2014	30-12-1899 16:00:00	Group D	Estadio Castelao	Fortaleza

e) Apply data blending between two sources

Steps:

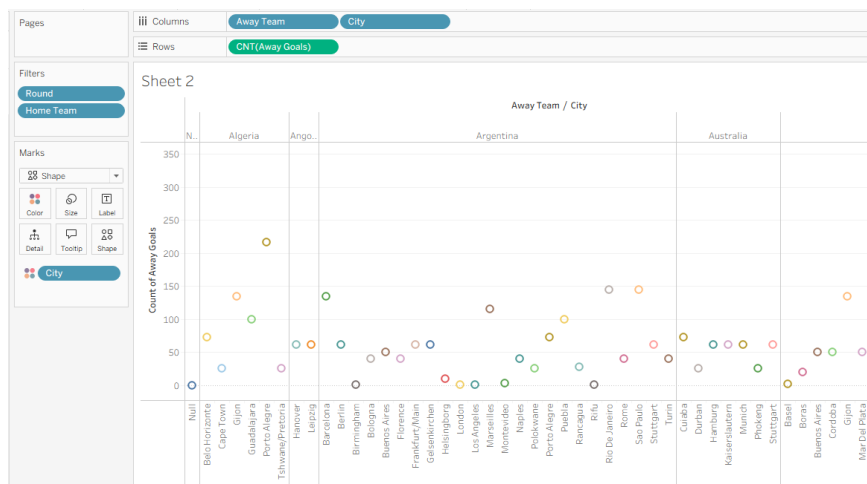
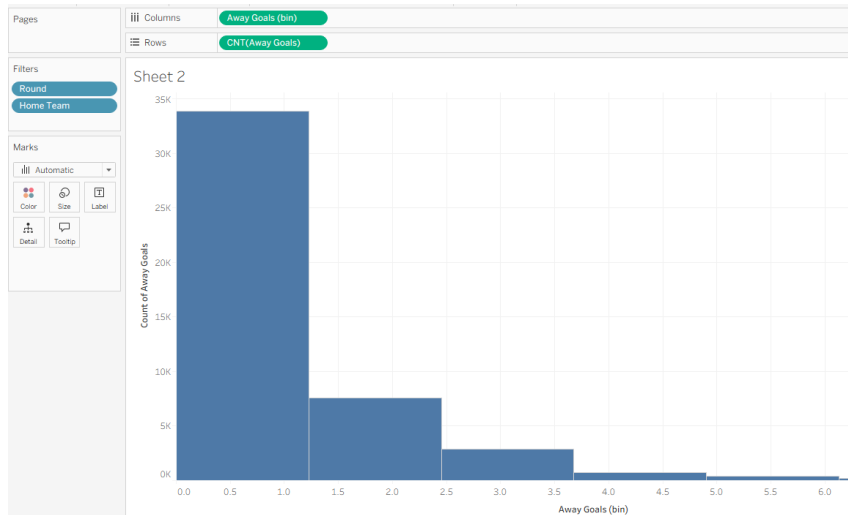
1. Connect to the first data source.
2. Add a second data source.
3. Use one as the primary source.
4. Drag fields from the secondary source.
5. Ensure the linking field matches.

The screenshot shows the 'Blend Relationships' dialog box. It has a 'Primary data source' dropdown set to 'world_cup_results'. The 'Secondary data source' dropdown is set to 'Retail_sales - Retail_sales'. The 'Automatic' radio button is selected. The 'City' field is selected in the 'Secondary data source' list. The 'Day of the Week' field is selected in the 'Primary data source' list. The 'Add...' button is highlighted.

f) Implement different types of filters (dimension, measure, date)

Steps:

1. Drag a field to the **Filters** shelf.
2. Select the filter type:
 - Dimension
 - Measure
 - Date
3. Configure the filter options.

4. Click **OK**.g) Optimize performance using extracts and filtering
Steps:

1. Create an Extract instead of a Live connection for large datasets.
2. Apply filters early in the data source.
3. Hide unused fields.
4. Use context filters where appropriate.
5. Remove unused worksheets and calculations.

world_cup_results

Connection
☒ Live ☐ ExtractFilters
1 | Edit

world_cup_results...

world_cup_results.csv 24 fields 288 rows

Name	world_cup_results.csv	Description	No description available.
Type	Field Name	Physical Table	Remote ...
Year	Year	world_cup_results.csv	Year
Date	Date	world_cup_results.csv	Date
Time	Time	world_cup_results.csv	Time
Round	Round	world_cup_results.csv	Round

Year	Date	Time	Round	Stadium	City	Country
2014	16-06-2014	30-12-1899 16:00:00	Group F	Arena da Baixada	Curitiba	Brazil
2014	20-06-2014	30-12-1899 19:00:00	Group E	Arena da Baixada	Curitiba	Brazil
2014	23-06-2014	30-12-1899 13:00:00	Group B	Arena da Baixada	Curitiba	Brazil
2014	26-06-2014	30-12-1899 17:00:00	Group H	Arena da Baixada	Curitiba	Brazil
2014	16-06-2014	30-12-1899 16:00:00	Group F	Arena da Baixada	Curitiba	Brazil
2014	20-06-2014	30-12-1899 19:00:00	Group E	Arena da Baixada	Curitiba	Brazil
2014	23-06-2014	30-12-1899 13:00:00	Group B	Arena da Baixada	Curitiba	Brazil